PYTHON ACTIVITY-BASED LEARNING

Case Study 10: Multi-Criteria Internship Candidate Ranking

Dr. RVR NRI Institute of Technology | Python Programming

Student Name: _____ **Roll No:** _____

Branch / Section: _____

Problem Statement

Evaluate internship candidates using academic marks, assessment scores and attendance, then rank qualified candidates.

Objectives

• Apply minimum eligibility criteria.

• Calculate a combined score.

• Rank eligible candidates using a tie-break rule.

Assumptions

• Marks, assessment and attendance are percentages out of 100.

• Eligibility requires marks of at least 60, assessment of at least 50 and attendance of at least 75.

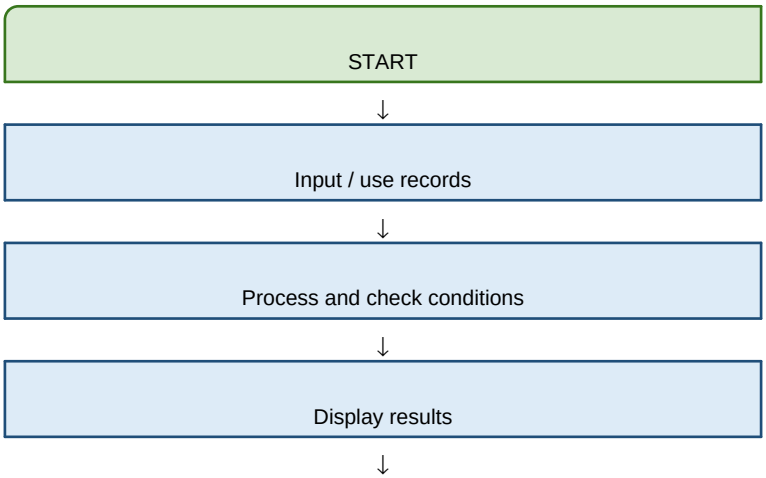
• Final score is 60% marks plus 30% assessment plus 10% attendance.

• Ties are resolved by higher assessment score.

Algorithm

1. Store candidate tuples with name, marks, assessment and attendance.
2. Check all three eligibility conditions.
3. Calculate weighted score for eligible candidates.
4. Store eligible candidates with their score and assessment.
5. Sort by weighted score and assessment in descending order.
6. Display the ranked candidates.

Flowchart



STOP

Pseudocode

```
START
STORE candidate tuples
SET eligible list empty
FOR each candidate
    CHECK marks >= 60, assessment >= 50, attendance >= 75
    IF eligible
        CALCULATE weighted score
        ADD candidate and score to eligible list
SORT by score then assessment, descending
DISPLAY ranked list
STOP
```

Python Source Code

```
candidates = [
    ("Asha", 85, 80, 90),
    ("Bharat", 72, 75, 80),
    ("Charan", 90, 45, 95),
    ("Divya", 78, 88, 85)
]

eligible = []

for name, marks, test, attendance in candidates:
    if marks >= 60 and test >= 50 and attendance >= 75:
        score = marks * 0.60 + test * 0.30 + attendance * 0.10
        eligible.append((name, score, test))

eligible.sort(key=lambda x: (x[1], x[2]), reverse=True)

print("Rank  Candidate  Score")
for rank, item in enumerate(eligible, start=1):
    print(rank, item[0], round(item[1], 2))
```

Test Cases

Test input / condition	Expected result
Asha: 85,80,90	Eligible; score 84.5
Charan assessment 45	Not eligible
Divya: 78,88,85	Eligible; score 81.9

Output Screenshot

Sample console output (formatted to show the expected execution result):

```
Rank  Candidate  Score
1 Asha 84.5
2 Divya 81.9
3 Bharat 73.5
```

Conclusion

The program shortlists candidates and ranks them using weighted scores and a deterministic tie-break rule.